

Problem Set #5

Due: Friday, April 24, 11:59pm

You may work with your study group on this assignment. However, even if you work with others you must write up and submit your own assignment. Late assignments will not be accepted.

1. Answer the questions for the Lab: Exploring the Consumer Expenditure Survey.
2. Suppose $X_1, X_2, \dots, X_n$ is an i.i.d. random sample, with $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$, and with $n \geq 2$. Consider the following four estimators of μ : $W_1 = \bar{X}$, $W_2 = \bar{X}/2$, $W_3 = (X_1 + X_n)/2$ and $W_4 = \frac{n-1}{n} \bar{X}$.
 - (a) Calculate $E(W_j)$ for $j = 1, 2, 3, 4$. Which of the four estimators are unbiased? Of the biased estimators, which are asymptotically unbiased?
 - (b) Calculate $\text{Var}(W_j)$ for $j = 1, 2, 3, 4$. For which estimators does the variance go to zero as n increases without bound?
 - (c) Which of the estimators is consistent?
3. Show that $\frac{1}{n+1} \sum X_i$ is a consistent estimator of $E[X_i]$.
4. A *Wall Street Journal* (January 4, 1991) article reported that “The average American consults with a doctor 5.4 times a year. Most contacts are at the doctor’s office, an average of 3.2 visits per person each year.” To test whether the average is 3.2 visits at the doctor’s office, a local group of physicians pooled data on the average number of visits for those who actually came to their offices and concluded that their sample average was 6.5 visits a person each year, well above the claimed average of 3.2 visits. Comment on their sample design and the validity of their conclusion.
5. (Vacation Travel) An airline does a market research survey on travel patterns. It takes a simple random sample of 225 people aged 18 and over in a certain city, and works out the 95% confidence interval for the average distance they traveled on vacations in the previous year. This was 488 to 592 miles. Say whether each statement below is true or false; give reasons. If there is not enough information to decide, explain what else you need to know.

- (a) The average of the 225 distances is about 540 miles.
 - (b) The standard deviation of the 225 distances is about 390 miles.
 - (c) The histogram for the 225 distances is close to the normal curve.
 - (d) The probability distribution for the sample average is close to the normal curve.
 - (e) A 95% confidence interval based on a sample of 450 people will be about half as wide as one based on a sample of 225 people.
6. (Portfolio value prediction intervals.) Les Izmor is an employee of Epsilon Inc, a major software firm. Since joining Epsilon nearly ten years ago, Les has set aside a portion of his income as savings for retirement. Like many employees, Les used these savings to purchase stock in Epsilon each month, and the total value of his Epsilon stock is now \$1,000,000.
- (a) The quarterly rate of return on Epsilon stock is approximately normally distributed with mean 6.37% and standard deviation 18.08%. Determine an interval that, with probability 0.9, will contain the (total) value of his stock in Epsilon one quarter from now (Note: The interval should be symmetric, meaning that the possible outcomes above the interval have 1/2 the remaining probability, or 0.05, and the outcomes below the interval have the other 1/2 of the remaining probability, 0.05). This interval is known as a 90% prediction interval for Les' portfolio value in one quarter. For simplicity in calculating the intervals for this problem, you should assume that Les makes no additional purchases of Epsilon stock during this quarter.
 - (b) Last summer, Les took a course on retirement planning where he learned about the value of diversification. He now wants to diversify his holdings between the Epsilon stock and the Sigma fund, a high-risk, high-return mutual fund. He knows that the quarterly rate-of-return on the Sigma fund is approximately 10.31% with standard deviation 22.85%. In addition, the correlation between the quarterly return on Epsilon stock and the quarterly return on the Sigma fund is 0.251. Assume that the return on Epsilon stock and the return on the Sigma fund are jointly normally distributed.
Suppose that Les keeps 46% of his current \$1,000,000 retirement savings in Epsilon stock, and puts the remaining 54% in the Sigma fund. Find a (symmetric) 90% prediction interval for the value of this portfolio one quarter from now.
7. (Party ID) The town of Hayward (CA) has about 50,000 registered voters. A political research firm takes a simple random sample of 500 of these voters.

In the sample, the breakdown by party affiliation is 115 Republicans, 331 Democrats, and 54 Independents. Calculate a 98% confidence interval for the true percentage of Independents among Hayward's 50,000 registered voters.

8. In a *Wall Street Journal* (August 17, 1987) article on selling machine tools to U.S. firms, Toyota representatives claimed that "Toyota machines would operate 95% of the time." In 47 randomly selected eight-hour shifts, it was found that the Toyota machines were down because of breakage an average of a half hour per shift, with a standard deviation of ten minutes. Do these observations lead you to reject the Toyota claim?
9. Answer the questions for the Lab: Testing Differences in Expenditures.
10. (Crime) In the U.S., there are two sources of national statistics on crime rates: (i) the FBI's Uniform Crime Reporting Program, which publishes summaries on all crimes reported to police agencies in jurisdictions covering virtually 100% of the U.S.; (ii) the National Crime Survey (NCS), which is based on interviews with a nationwide probability sample of households. In 1992, 4.9% of the households in the NCS sample told the interviewers they had experienced a burglary within the past 12 months. The same year, the FBI reported a burglary rate of 32 per 1,000 households, or 3.2%, for the U.S. as a whole.

If the FBI's 3.2% figure is the true burglary rate for the population of U.S. households in 1992, what is the probability that the National Crime Survey would obtain a result of 4.9% or more? You may assume that the NCS is based on a simple random sample of 50,000 households out of a population of 100 million households. What do you conclude?
11. (Tax fraud) Each year, the Internal Revenue Service attempts to estimate total tax compliance: whether Federal tax revenues are above or below what they should be, and by how much. For many years, this was performed by the IRS Tax Compliance Measurement Program (TCMP). To conduct the TCMP, the IRS drew a simple random sample of U.S. citizens and audited each one. In a TCMP audit, the true tax liability of the individual (or household, if the individual files Federal income taxes jointly with a spouse) is assessed by the IRS after detailed examination of the individual's (or household's) total income by IRS agents. The tax liability determined in the audit is then compared to the actual tax return filed by the individual or household. The IRS extrapolates from the average tax underpayment in the TCMP sample to estimate the total personal income tax underpayment level in the U.S. Until recently, TCMP audits were performed on approximately 150 thousand randomly selected taxpayers annually.

A Congressional representative asserts that tax fraud is on the rise in the U.S. To examine the issue, the IRS provides the following (hypothetical) sample data on tax underpayment amounts from the TCMP survey for two consecutive years.

	Year 1	Year 2
Sample Average	\$423.27	\$426.84
Sample SD	\$359.21	\$361.14
Sample Size	151,627	153,249

Some taxpayers underpay their taxes (a positive number), while others overpay their taxes (a negative number); the average of these numbers for all taxpayers in the survey is listed in the first row of the table and the SD of these numbers is listed in the second row. Assume the numbers have been adjusted for inflation.

- (a) Do the data indicate that tax fraud is on the rise?
 - (b) Why is such a large sample used in the TCMP?
12. The U.S. Supreme Court in *International Brotherhood of Teamsters v. United States* (1977) stated, "It is ordinarily to be expected that nondiscriminatory hiring practices will in time result in a work force more or less representative of the racial and ethnic composition of the population in the community from which employees are hired." In one area, 25 percent of the accountants available for new accounting jobs are African American. Thus, a company that has not discriminated against African American applicants should hire a proportion of African Americans that is not significantly different from 0.25. A company in this area hired 80 new accountants over the past several years, 16 of whom were African American. Does this indicate discriminatory hiring?